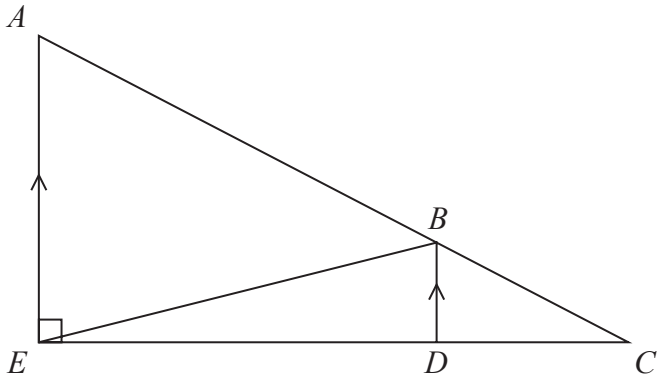
20 AEC is a right-angled triangle.
B is a point on AC and D is a point on EC.



The area of triangle BCD is $5\,cm^2$
The area of triangle EBD is $15\,cm^2$

Work out the area of triangle ABE.
You must show all your working.

Triangles EBD and BCD have the same height from B.

ED : DC = 15 : 5 = 3 : 1

1 mark
Equal heights: areas follow the bases.

EC : DC = 4 : 1

Triangles AEC and BDC are similar (AA), with AE parallel BD.

1 mark
The length scale factor from BDC to AEC is 4.

Area AEC = $4^2 \times 5 = 80\ cm^2$

1 mark
Square the length scale for area.

Area ABE = 80 – 15 – 5 = 60 cm^2 ..... 60 ..... cm^2

1 mark
Subtract BOTH smaller triangles.

(Total for Question 20 is 4 marks)

DO NOT WRITE IN THIS AREA
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